

Exp. 4 Setting up a Local Area Network. (LAN)

Date: 31/8/26

Aim:-

TO design, set up, and configure a LAN connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource / data sharing across the network.

Equipment Required:-

S.No	Equipment	Quantity	Specification.
1	Personal computers	4	Ethernet (NIC) enabled.
2	Network Switch	1	8/16/24 port, unmanaged / managed.
3	Ethernet cables (cat 5e / cat 6)	4	Straight-through, RJ-45 connectors.

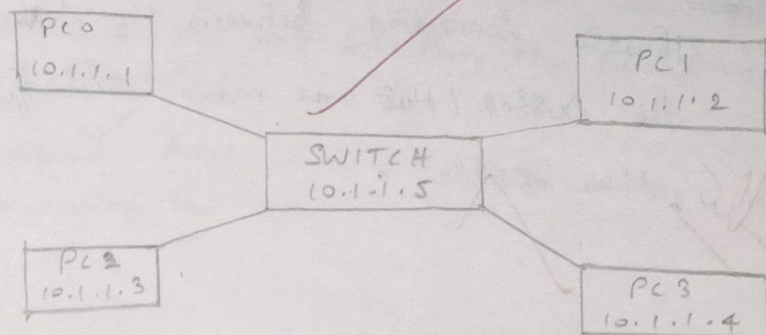
Theory:-

A LAN connects computers and devices within a limited area to share resources like files, printers, and Internet access.

In this experiment, a Star topology is used, where all devices are connected to a central switch. The switch operates at OSI layer 2, forwarding data based on MAC addresses. All devices must have IP within the same network range to communicate each other.

Network Diagram:

LAN was implemented using a Star topology as shown below, with 4pc individually to a central switch.



5. IP Address Configuration table.

Device	IP Address	Subnet Mask	Connected to.
PC0	10.1.1.1	255.0.0.0	Switch port Fa0/11
PC1	10.1.1.2	255.0.0.0	switch port Fa0/12
PC2	10.1.1.3	255.0.0.0	switch port Fa0/13
PC3	10.1.1.4	255.0.0.0	switch port Fa0/14
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface.

6. Implementation in Cisco Packet Tracer.

Step 1:- Place the Devices.

1. Open Cisco Packet Tracer.
2. Drag 4 PCs and 1 switch to the workspace.

Step 2: connect the Devices.

3. Select Copper Straight-Through cable.
4. Connect each PC to the switch using Fa0/11 - Fa0/14 ports.
5. Ensure the link lights turn green.

Step 3: Assign IP Addresses.

6. Open PC-Desktop → IP configuration.
7. Assign the static IP Address and Subnet Mask to each PC as per the IP table.

Step 4:- Configure Switch Management IP.

8. Open Switch CLI and enter:

enable

configure terminal.

Interface vlan 1

ip address 10.1.1.5 255.0.0.0

no shutdown

exit.

Step 5:- Verify Connectivity

9. Open PC → Desktop → Command Prompt.
10. Use ping (e.g., ping 10.1.1.2) to test communication.
11. Reply from... Confirms Successful connectivity.

Step 6:- Demonstrate Data Sharing.

12. Add a server -PT and connect it to the switch.
13. Assign IP: 10.1.1.10 and subnet mask: 255.0.0.0.
14. Enable the HTTP service under Services.
15. From another PC, open Desktop → Web Browser, enter the server IP, and verify the webpage loads.

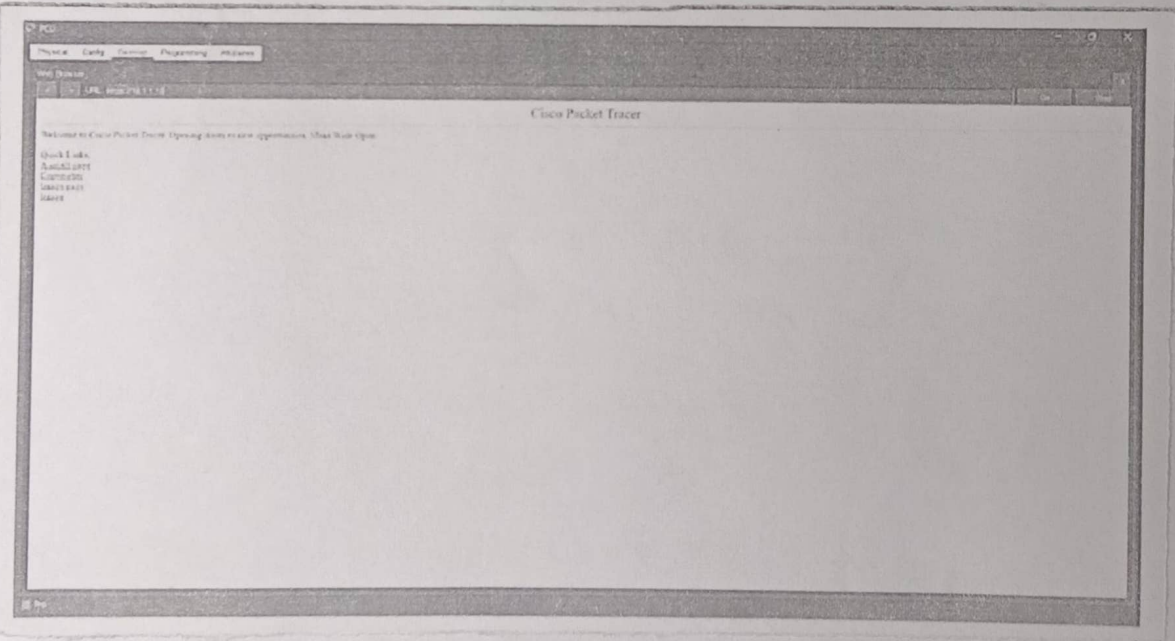
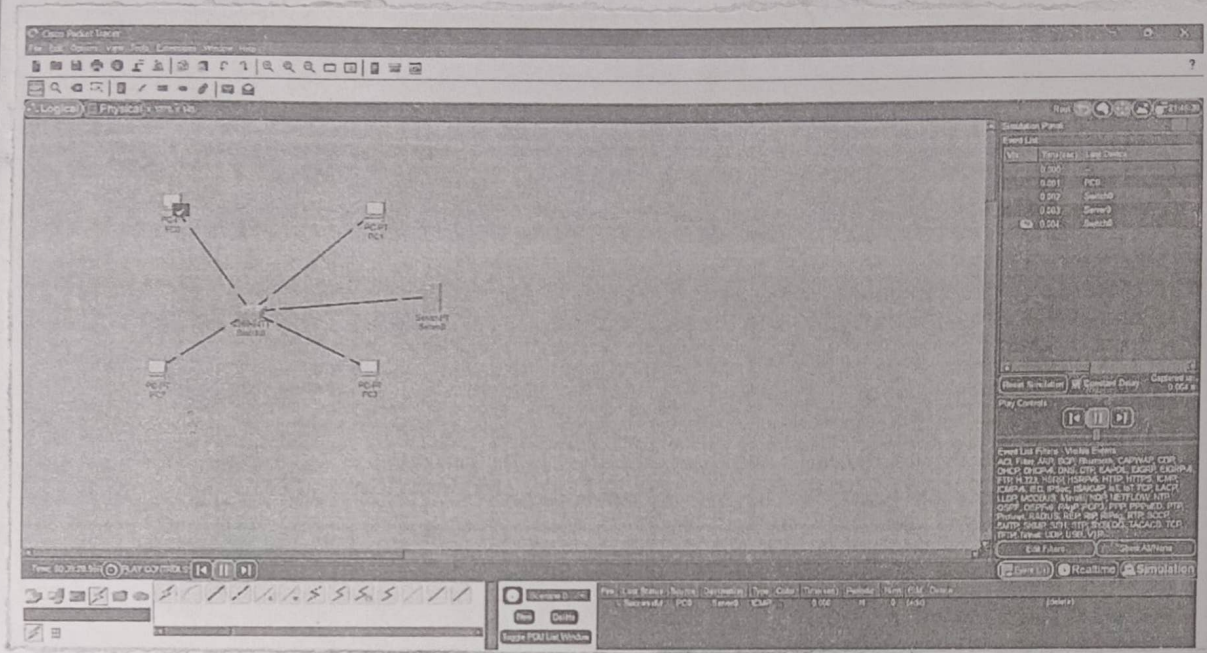
Step 7: Observe Packet Flow.

16. Switch to simulation mode.
17. Send a ping or web request and use Play/Capture to observe packet forwarding through the switch.

7. Observation :-

All four PCs were connected to the switch in a star topology and assigned unique static IP addresses in the same subnet (10.0.0.0/255.0.0.0).

Ping tests were successful, confirming network connectivity. The shared (server / resource) was also accessed successfully, verifying proper LAN communication and resource sharing.



Result:-

Thus, the LAN was setup and the webpage was shown successfully.